Scheduler Rankings and Performance Trade-offs in a Common-Model LLM-Serving Evaluation

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A scheduler ranking is a property of an evaluation contract as well as a workload. Comparing implementations with different service progression, resource enforcement or policy information can obscure what a ranking measures. We evaluate four project-maintained LLM-serving scheduler adaptations under one explicit CPU discrete-event model. The complete finite-corpus design contains 120 windows from three trace sources, six inherited arrival/deadline settings and two KV reservation capacities: 5,760 validated cells, including 144 reused pilot cells. All requests complete. Across six source/capacity aggregates, the SLO-gated adaptation reduces mean latency by 11.3–34.0% relative to chunked-FCFS while reducing token throughput by 1.65–4.01%; BurstGPT's mean window-level p95 latency increases at both capacities. The urgency gate never binds, so these gains do not demonstrate urgency-aware scheduling. Chunked-FCFS and decode-first are exactly equivalent in all 1,440 matched conditions. There are no strict 10% practical reversals between source means, although window-level and metric-dependent reversals occur. A separate 134,000-study known-truth diagnostic identifies calibration concerns in historical inference procedures; it does not confer statistical guarantees on trace comparisons. The contribution is an auditable model-based evaluation showing how absolute trade-offs, structural equivalences and aggregation jointly constrain ranking claims, rather than a new scheduler or a native-GPU performance prediction.

CCS Concepts: • **Computing methodologies** → **Simulation evaluation**; • **Computer systems organization** → *Cloud computing*.

Additional Key Words and Phrases: LLM serving, scheduling, performance evaluation, discrete-event simulation, workload dependence

1 Introduction

LLM serving must reconcile prompt processing, incremental token generation and retained key–value (KV) state. Batching and memory-management choices change the feasible schedule; workload lengths and the chosen service objective change which schedule is preferable [2, 11, 19]. A comparison therefore needs more than a shared trace and a list of scheduler names. It needs an execution contract specifying what work a decision causes, how resources are reserved and what information is available when the decision is made.

This issue arose in our historical LLM Serving Scheduler Portability (LSSP) comparison. Its saved implementations combined different prefill progression and KV-enforcement rules. Replaying those implementations reproduced their outcomes, but did not isolate scheduling under a common resource model. Moreover, practical ranking changes were filtered through an inferential procedure whose calibration had not been established. These are separate problems: harmonizing execution does not validate an inference procedure, and reproducing its arithmetic does not harmonize execution.

We address the comparison problem with CM-0.1, a deliberately simple, fully specified discrete-event contract. Four adaptations share the same service clock, work budget, reservation rule, admissible actions and online information. Independent event checks reconstruct execution from journals rather than trusting the engine's summaries. We then evaluate the complete frozen corpus,

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retaining every condition and reporting absolute metrics before rankings. The analysis describes this corpus; requests and repeated settings do not become independent observations.

Workload-sensitive serving evaluation is established [14], as are predictive serving simulators [1, 7]. Our contribution is a different, bounded empirical object: the joint behavior of four identified adaptations under a fixed contract, with matched-window comparisons, explicit practical-reversal denominators and an account of which ranking distinctions the model can express. CM-0.1 uses a fixed batch cost rather than fitted GPU timings. It supports model-based comparison, not a claim to greater predictive accuracy than those simulators.

The complete evaluation is informative partly because it limits the original portability narrative. Source-mean practical reversals are absent. Two adaptations execute equivalently. The nominally SLO-gated adaptation improves mean latency while losing throughput and sometimes tail performance; its urgency gate is inactive. Window and metric changes still produce practical reversals, so aggregate stability does not imply uniform performance. A separate known-truth study shows why the earlier inference pipeline cannot turn these descriptive comparisons into general calibration claims.

We make four contributions:

(1) An auditable execution, resource and information contract, with explicit adaptation identities and independent event conformance checks.
(2) A complete 5,760-cell evaluation over 120 workload windows, six inherited arrival/deadline settings and two reservation capacities, including absolute performance and matched-window trade-offs.
(3) A descriptive account of ranking stability that retains exact ties, structural equivalences, numerical near-ties and strict practical reversals at distinct aggregation levels.
(4) A scoped interpretation of historical comparisons, informed by a 134,000-study calibration diagnostic that separates implementation fidelity from mathematical definitions and statistical validity.

Three questions organize the paper. **RQ1:** How do the absolute performances of four adaptations vary across workload windows, inherited arrival/deadline settings and KV reservation capacities? **RQ2:** How do workload variation, aggregation, metric selection, ties and practical differences affect scheduler rankings? **RQ3:** What do the historical comparison audit and known-truth calibration establish about the reliability and limitations of these comparisons? Sections 5–7 answer them in that order.

2 Background and Related Comparisons

Serving mechanisms. Orca established iteration-level scheduling for generative inference; vLLM's PagedAttention addresses KV-memory management; Sarathi-Serve uses chunked prefill to manage throughput–latency trade-offs [2, 11, 19]. These systems motivate our adaptation lineage, but their hardware, kernels and allocators are not executed here. The distinction is especially important for SLAI, whose published scheduling framework includes theory and a richer practical policy [4]. Our restricted prompt-ordering and urgency-gate adaptation neither implements that complete policy nor inherits its guarantees. Prefill/decode disaggregation in DistServe and dynamic migration in Llumnix illustrate further mechanisms outside the single-device, nonpreemptive CM-0.1 action space [16, 21].

Workload-sensitive evaluation. Papaioannou and Doudali explicitly examine how dataset and inference scenario affect serving-system evaluation [14]. Thus workload dependence itself is not our novelty. BurstGPT supplies production-derived traffic [17]; ServeGen studies workload characterization and generation [18]. Our experiment instead freezes existing windows and jointly

reports performance under specified resource and metric axes. It does not evaluate a workload generator or claim that three sources represent all serving traffic. Work on trace resampling for parallel schedulers likewise shows why workload structure matters [20]. We retain chronology descriptively rather than borrowing an independence assumption from a different resampling design.

Simulation and validation. Vidur combines profiling and predictive models to estimate serving performance [1]. LLMServingSim composes hardware/software simulation at iteration granularity [7]. Their hardware-prediction purpose differs from our fixed-cost comparison contract. Following the distinction between model verification and validation [15], we verify that schedules conform to CM-0.1; this does not establish operational agreement with a GPU system. The useful question here is which comparative statements survive explicit work, resource and information accounting, and which follow from the abstraction itself.

Ranking and inference. Comparing algorithms across datasets is well established [8]. A rank, a maximizing set, a practical pairwise reversal and an absolute performance difference answer different questions. We do not propose a new rank statistic. Likewise, BH, BY and Holm are established multiple-testing procedures [5, 6, 10]; their error-control arguments require appropriate component p-values and assumptions. Section 7 tests a specific historical pipeline and separate comparison procedures, rather than presenting multiplicity adjustment as a remedy for an unidentified estimand.

3 The Common-Model Comparison Contract

3.1 Execution, resources and information

CM-0.1 models one abstract device. A nonempty batch lasts exactly $\Delta = 0.001$ simulated seconds and may execute at most 32,768 work tokens. Each prefill token costs one work unit; each scheduled decoding request costs one unit and emits one token. A request's final prefill batch emits no output, so decoding starts no earlier than the next batch. Requests arriving during a batch become eligible at its end. When no work is available, the clock advances directly to the next arrival.

Every resident reserves a full 32,768-token context, regardless of its unknown output length. With capacity K, simultaneous residency is bounded by $\min(64, \lfloor K/32768 \rfloor)$. The two capacities, 131,072 and 262,144 token slots, therefore permit *four and eight* residents, respectively. Used KV is tracked separately from executed prompt and output prefixes, rounded up to 16-token blocks. All frozen prompt-plus-output lengths fit the context bound (maximum 30,053). Reservation avoids output prediction and admission deadlock, at the cost of conservative concurrency. It is not an incremental native KV allocator.

A policy sees only arrived request IDs and stable ordinals, arrival times, prompt lengths, acknowledged progress, resident status, the last-output or prefill-completion time, and the current clock. Actual or remaining output length, source identity, future arrivals and output-derived deadlines are hidden. The engine receives explicit per-request work spans and either executes that plan exactly or rejects it. No eviction, preemption, migration, prefix sharing or recomputation is available. Completion releases the reservation. The common administrative horizon is the last arrival plus 50 seconds; unfinished requests would remain explicitly unfinished, not become observed latency samples.

3.2 Why some mechanisms coincide

The four rules in Table 1 have distinct identities, but need not induce distinct executions. At most eight residents or newly admitted requests can receive work in a batch. With a 512-token prefill

Table 1. Adaptations under CM-0.1. Names below denote project-maintained rules, not executions of the original systems. All rows share the contract in Section 3.

Adaptation / lineage	Executed scheduling rule
Whole-FCFS / vLLM-inspired	Admit FCFS whole prompts when a slot and budget permit; defer resident decodes in such batches. Otherwise decode residents.
Chunked-FCFS / chunked vLLM-inspired	Resident work before waiting work, stable arrival order; prefill cap 512 tokens per request per batch.
Decode-first / Sarathi-inspired	Resident decodes, resident prefills, then waiting prefills, each class FCFS; the same 512-token prefill cap.
SLO-gated / SLAI-inspired	Critical resident decodes first, then resident prefills, waiting prefills by prompt length and ordinal, then remaining decodes. Critical means time since last output/prefill completion at least 0.495 seconds. The prefill cap is 512 tokens.

cap, the three chunked rules use at most $8 \times 512 = 4096$ work tokens, well below the 32,768 budget. Reordering resident decode and prefill work therefore does not exclude any of it, and the fixed batch cost is independent of order. Chunked-FCFS and decode-first admit waiting requests in the same order and advance the same prefixes. Their equality is a structural consequence in this domain, verified against all final records.

The same budget slack lets SLO-gated execute all eligible residual decodes in every batch. The time since an acknowledged output or completed prefill cannot accumulate to its 0.495-second gate. The gate is inactive throughout these executions; prompt-length ordering of new admissions is the relevant distinction. A gain attributed to an active urgency gate, or to full SLAI, would misidentify the executed mechanism. Whole-FCFS can differ because whole-prefill batches explicitly defer decodes and its larger prompt spans can constrain admission within the budget.

3.3 Conformance evidence

An independent event referee reconstructs request progress, intent versus executed spans, eligibility, token work, reservations, block-rounded KV, timestamps, completion and metric accounting without importing the engine or policy implementation. The execution gate also tested deterministic replay, validated restart, input identity and future/latent-output noninterference. All 5,760 journals passed the exhaustive in-allocation check before final analysis. The subsequent review independently checked all result and input identities, reran the separate summary verifier and replayed 24 structurally selected journals. Those fresh sampled checks supplement, rather than replace or purport to repeat, the exhaustive pass. This evidence establishes conformance to the contract, not physical timing accuracy or security isolation of policy code.

4 Corpus, Metrics and Analysis Design

4.1 Frozen inputs and provenance

The corpus comprises 40 windows of 200 requests from each of Azure 2024, Bailian/Qwen and BurstGPT: 120 windows and 24,000 underlying request records. Azure's official release, Alibaba's pinned usage traces and the BurstGPT dataset have distinct release provenance [3, 13, 17]. BurstGPT traffic also originates on Azure infrastructure; the sources are not three independent providers. Source-reported lengths are not common-tokenizer measurements, and conversation/context semantics need not coincide. The Bailian adapter reads `input_length` and `output_length` directly

as token counts; the release's 16-token block size describes prefix hashes, not a length conversion. Synthetic priorities and deadlines are benchmark attributes, not observed service commitments.

Each source's 40 windows includes ten reused from the earlier historical pilot and thirty from a pinned extension sampler. Selection of the extension uses valid-row ranges and fixed seeds rather than scheduler outcomes. Historical pilot outcomes nevertheless preceded the expanded corpus and motivated subsequent work; this is not a new random confirmatory sample. The 144 reused C01 execution cells are a different reuse unit: one window per source, six settings, four adaptations and two capacities. They passed exact input, code, configuration and journal-identity checks before byte-for-byte reuse. C01-FULL adds 5,616 cells without changing the pilot's execution semantics.

4.2 Inherited arrival/deadline settings

For window w and inherited setting r, the frozen transformation uses $\lambda(w, r) = c_r \lambda_{\mathrm{FIFO}}(w)$, with the multipliers $c_r = 0.5, 0.8, 1.0, 1.1, 1.2, 1.5$. The labels LOW, PRE_KNEE, KNEE, POST_KNEE, OVERLOAD and HIGH_PRESSURE come from the earlier FIFO calibration. They are not newly measured saturation regions for CM-0.1, nor common absolute arrival rates across windows. Absolute factors range from 22.525–180.873 (Azure), 17.956–186.580 (Bailian) and 102.884–725.020 (BurstGPT).

Crucially, the frozen transformation scales both arrival gaps and synthetic deadline slack. If the unscaled arrival/deadline pair is (a_i, d_i) and the first arrival is a_0, the transformed pair is

$$a'_i = (a_i - a_0)/\lambda, \qquad d'_i = a'_i + \max(0, d_i - a_i)/\lambda. \tag{1}$$

Token lengths and priorities are unchanged. Thus comparisons across these settings describe joint arrival/deadline stress. In particular, ANWG changes do not identify an isolated arrival-load effect. The deadlines retain an output-derived historical construction in the metric layer; none is revealed to a policy. No request is truncated to make the design fit.

4.3 Metrics and support

Let a_i, f_i and c_i denote arrival, first-output and completion times, and o_i output length. Completed-request latency is $c_i - a_i$; TTFT is $f_i - a_i$; TPOT is $(c_i - f_i)/(o_i - 1)$ only for $o_i > 1$. We report the mean and p95 of each population. The p95 linearly interpolates sorted values at position $0.95(n-1)$, using zero-based indexing. An average of window p95s is always named a *mean window-level p95*; it is not a pooled request percentile.

Completed-output token throughput is $\sum_{i \in C} o_i/T$, emitted-token throughput counts all executed output work over T, and request throughput is $|C|/T$, where T runs from first arrival to final completion, or to the administrative horizon when unfinished. Completion fraction is $|C|/200$. With positive priority weights v_i, arrival-normalized weighted goodput is

$$\mathrm{ANWG} = \frac{\sum_{i \in C} v_i \, \mathbf{1}\{c_i \le d_i\}}{\sum_{i \in A} v_i}. \tag{2}$$

The frozen corpus uses uniform priorities, so ANWG here is the fraction of arrivals completed by their synthetic deadlines. These five larger-is-better endpoints and six smaller-is-better latency endpoints form eleven ranking metrics. Counts, busy-device time, executed work, observation time and used/reserved KV are retained as accounting measures. Undefined values would remain missing with their support counts, not be imputed as zero.

4.4 Analysis unit and comparison rules

The whole workload window is the principal unit. Settings, capacities and adaptations are repeated conditions on it. Source means at a fixed setting and capacity weight all 40 windows equally.

Table 2. Input and execution provenance. Full SHA-256 values, per-window identities and source filenames are retained in the accompanying manifest; short prefixes here are navigational. Release-specific redistribution permissions and reviewer access remain author-review items.

Object	Frozen identity and transformation
Azure 2024	Official 2024 inference trace; arrival and source-reported prompt/output lengths; 40 windows.
Bailian/Qwen	Upstream revision `5f7439c51ec2`; source token lengths read directly by the pinned adapter; 40 windows.
BurstGPT	`BurstGPT_without_fails_2.csv`; source hash `56193aa9b2bb`; 40 windows from the released successful-request file.
Corpus	Physical hash `97fbaf6a4b9b`; canonical window hash `0d1aa06ccbee`; exact per-window source hashes and row positions retained.
C01-FULL	Execution commit `2d8342e699f3`; pilot semantics `9aed0206de66`; 5,760 cells, 144 reused.
STAT01	Recovery execution `1ac4c8dd1421`; 112,275 reused and 21,725 newly generated studies, original streams preserved.

Source/capacity summaries first average six settings per window, then average windows equally. Matched differences subtract policy outcomes for the same window, setting and capacity; cross-source windows are not matched by ordinal. All statements are finite-corpus descriptions. No C01 confidence intervals, p-values or independent-window sampling assumptions are introduced.

Exact equality defines ties and average ranks; maximizing sets are retained rather than resolved to a single policy ID. A frozen secondary top-set sensitivity admits values within $10^{-12}\max(1, |\text{best}|)$ of the best. It does not replace exact rankings. For policies A, B, orient $D = x_A - x_B$ for a larger-is-better metric and $D = x_B - x_A$ for a smaller-is-better metric. The relative margin is $|x_A - x_B|/|x_{\text{loser}}|$. A practical reversal between two conditions requires opposite signs of D and margins *strictly* greater than $1/10$ on both sides. Comparisons use exact rational representations of the saved binary values; equality is excluded without an arbitrary tolerance. A missing or zero losing value is unestimable; a nonzero tie is eligible with zero margin.

Reversal tables distinguish within-source window pairs, cross-source window pairs, settings, capacities, metrics and source means. All six policy pairs remain, including the equivalent adaptations. These comparisons overlap and have different denominator constructions; their frequencies are neither independent trials nor a causal decomposition of source, setting and metric effects. Chronological sensitivities use frozen 20/10/10 early/middle/late windows per source and four ordered ten-window blocks. Blocks follow source-file hash, raw time, row position and window ID, not equal-duration sampling. Appendix A supplies complete tabular views and supplementary displays.

5 Absolute Performance and Operating Conditions

5.1 Completion and performance magnitudes

All 5,760 cells complete all 200 requests. There is no administrative-horizon censoring and no missing cell-level ranking endpoint. The 1,152,000 request executions are repeated conditions on the 24,000 underlying records, not independent observations. Individual TPOT excludes one-output-token requests, leaving 103–200 observations per cell. Cell mean latency ranges from 0.008314 to 3.893923 simulated seconds, p95 latency from 0.008560 to 7.300052 seconds, and output-token throughput from 332.489 to 7,766.685 tokens per simulated second. These magnitudes describe the fixed-cost model.

Figure 1 shows every source/setting/capacity combination, with equal-window means. Appendix Table 4 gives compact source/capacity aggregates. The SLO-gated adaptation reduces aggregate mean latency relative to chunked-FCFS in all six source/capacity groups, by 11.3–34.0%. Yet token throughput falls by 1.65–4.01% in those same groups. On BurstGPT, mean window-p95 latency increases by 3.72% with four residents and 18.97% with eight, despite better mean latency and ANWG. On Bailian with four residents, ANWG decreases by 3.12% despite 24.05% lower mean latency. Lower average response time therefore does not imply better tails, completion-by-deadline utility or throughput.

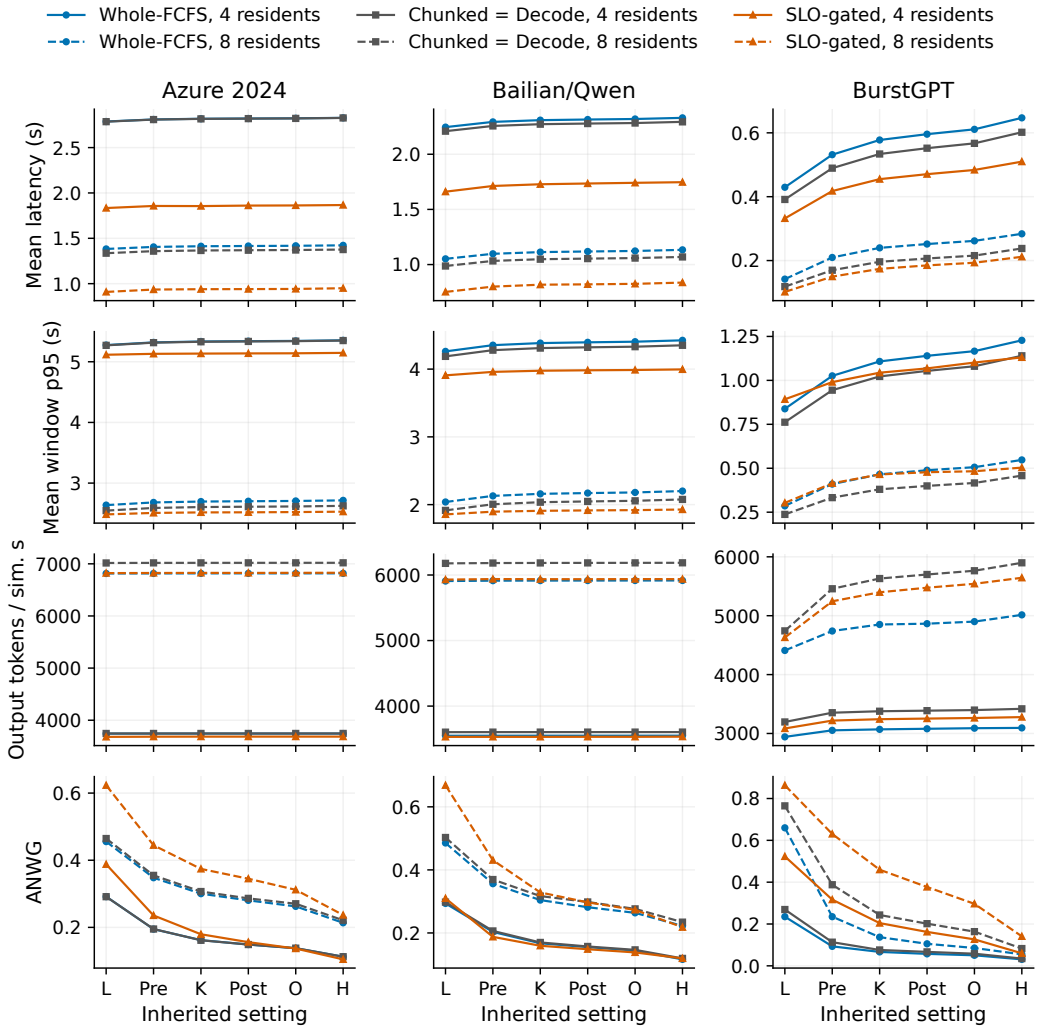


Fig. 1. Absolute performance: 40 equally weighted windows per point. L/Pre/K/Post/O/H denote the six inherited settings, not measured CM-0.1 saturation regimes. Solid/dashed lines allow four/eight reserved residents. Chunked-FCFS and decode-first coincide exactly. Seconds and throughput refer to simulated time; p95 panels average window-level p95s. No sampling confidence intervals are implied.

5.2 Matched-window trade-offs

Figure 2 retains all 40 windows per source and both capacities. Each point averages six *matched differences* within a window; it does not pool requests or substitute a difference between unmatched supports. The combination of lower mean latency and worse tail latency is visible on BurstGPT, while the size of the mean improvement varies appreciably by window and capacity. Throughput losses accompany many mean-latency improvements. Complete per-condition differences remain in the frozen analysis, and the plotted window summaries are supplied as CSV data.

SLO-gated is the unique mean-latency winner in 1,146 of 1,440 conditions, but the unique p95-latency winner in only 834 and the unique token-throughput winner in 482. Its unique ANWG wins number 1,004. The mean-latency count splits into 480/480 Azure, 468/480 Bailian and 198/480 BurstGPT conditions. These counts describe exact optima, not their practical margins. They complement the absolute differences rather than override the unfavorable throughput and tail outcomes.

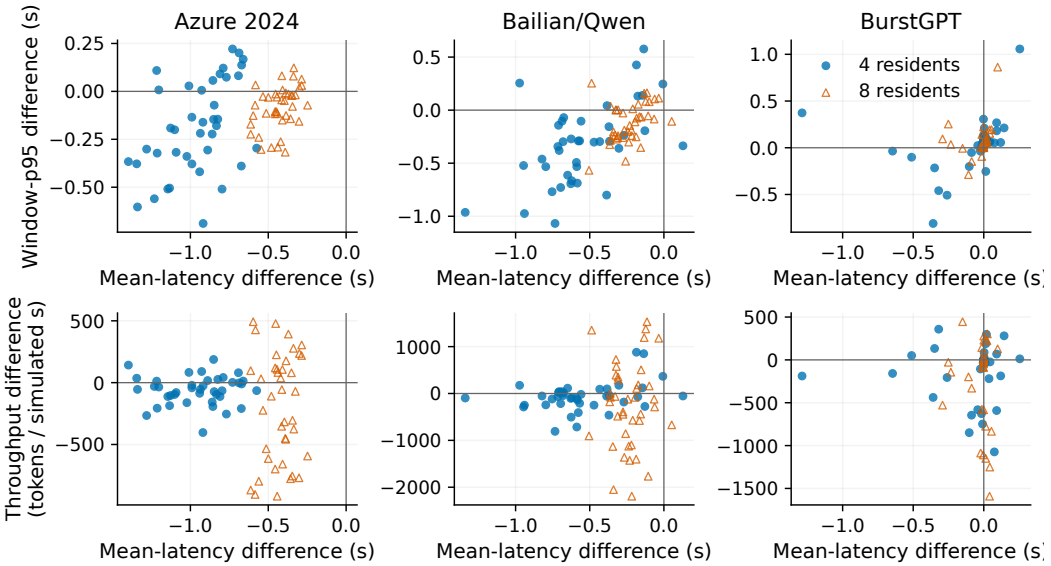


Fig. 2. SLO-gated minus chunked-FCFS for every window, averaged over its six matched settings. Each panel has 40 windows at each of two capacities. Leftward means lower mean latency; upward means worse window-p95 latency (top) or higher throughput (bottom). These are descriptive points with no independent-sampling interpretation.

5.3 Reservation and setting sensitivity

Doubling KV reservation capacity permits eight rather than four residents. At the source/capacity aggregate level, this improves throughput and lowers mean and mean-window-p95 latency for every adaptation. Actual used KV peaks at 47,872 tokens in the smaller-capacity runs and 62,880 in the larger ones, both below even the smaller nominal capacity. The observed constraint is full-context reservation-induced concurrency, not exhaustion of used KV at the nominal token limit. Paging, eviction and native memory-pressure benefits cannot be inferred from this comparison.

Inherited settings can change goodput substantially while leaving aggregate throughput almost unchanged. For Azure chunked-FCFS with four residents, LOW to HIGH_PRESSURE changes mean latency from 2.7858 to 2.8269 seconds and token throughput from 3,745.9 to 3,746.3, while ANWG

falls from 0.291375 to 0.112625. Equation 1 explains why interpreting this as an arrival-rate effect alone would be incorrect: the deadline criterion becomes tighter at the same time.

Answer to RQ1. Under this reservation-limited contract, the complete corpus exhibits a mean-latency/throughput trade-off with source-specific tail and goodput costs. Capacity changes the permitted concurrency; inherited settings jointly change arrivals and deadline slack. Completion is universal here, so the reported trade-offs do not result from selectively omitting unfinished requests. Their magnitudes are informative even where ranks remain unchanged.

6 Ranking Stability, Ties and Aggregation

6.1 Practical reversals depend on the comparison level

Table 3 retains every predeclared comparison axis and its eligible denominator. There are **zero source-mean strict-10% practical reversals**. Nevertheless, source rankings need not coincide: of 396 source-pair/setting/capacity/metric comparisons, 307 have identical exact top sets, 323 have intersecting top sets and 247 have identical full average-rank vectors. Differences below the practical threshold and changes in ties coexist with the null practical-reversal result.

Window heterogeneity is not erased by stable source means. For ANWG alone, within-source window comparisons produce 8,472 reversals among 162,900 eligible comparisons; cross-source windows produce 35,352/332,640. The latter is a Cartesian comparison of windows, not pairing by ordinal. ANWG setting and capacity comparisons yield 603/20,853 and 198/4,218, respectively. A corpus-average recommendation therefore does not describe every window or every utility. These are new adaptation/model estimands; they do not demonstrate persistence of the 36 historical-rule records.

The raw pair counts must be interpreted with their constructions. Within source, there are $\binom{40}{2} = 780$ window pairs per stratum; between sources there are $40^2 = 1600$. Settings generate 15 pairs, metrics 55, and every comparison retains six policy pairs. Including the equivalent chunked/decode-first pair preserves the frozen design, but its comparisons cannot be counted as additional independent behavioral evidence. Comparing these rates across axes is not a variance decomposition or a measure of causal importance.

6.2 Exact ties and metric redundancy

The exact top set is tied in 425/1,440 ANWG conditions, 289/1,440 mean-latency conditions, 586/1,440 p95-latency conditions and 906/1,440 token-throughput conditions. Completion is tied everywhere. With every request complete, emitted- and completed-token throughput coincide; request throughput differs by a workload-specific constant within each matched condition, so the three throughput rankings coincide as well. They are distinct recorded quantities, not three independent confirmations.

Tiny timestamp differences create another reporting issue. The chunked rules generate each resident output at essentially 1-ms intervals. The largest cell mean-TPOT difference between chunked-FCFS and SLO-gated is 1.80×10^{-16} seconds. The predeclared near-tie sensitivity changes the mean-TPOT maximizing set in 1,129/1,440 conditions and the p95-TPOT set in 104/1,440. Exact records are preserved; such distinctions have no practical performance interpretation. Appendix Table 5 reports all eleven endpoints.

Across 79,200 metric-pair ranking comparisons, exact top sets agree in 27,915 and intersect in 57,602. The difference between these counts shows why choosing one deterministic winner can obscure tied optima. Because these totals include redundant endpoints and numerical TPOT distinctions, the meaningful evidence is the absolute mean/tail/throughput trade-off, not the metric-pair disagreement count alone.

Table 3. Strict practical reversals across the complete panel. A reversal exceeds 10% relative to the losing value on both sides and changes sign. Eligible counts retain nonzero ties; zero losing values are unestimable. All comparisons are dependent descriptive enumerations.

Axis	Reversals	Eligible	Unestimable	Total
Within-source windows	34,394	1,847,700	5,580	1,853,280
Cross-source windows	109,913	3,788,640	12,960	3,801,600
Inherited settings	920	236,853	747	237,600
KV capacities	350	47,418	102	47,520
Metric choice	11,795	473,580	1,620	475,200
Source means	0	2,376	0	2,376

6.3 Chronological sensitivity

The four ten-window blocks preserve variation hidden by source-wide aggregation (Appendix Figure 4). SLO-gated minus chunked-FCFS ANWG block means range from −0.0120 to 0.2030 for Azure, −0.0300 to 0.2330 for Bailian and 0.0150 to 0.3275 for BurstGPT. Negative differences occur in 8/48, 24/48 and 0/48 block/setting/capacity combinations, respectively. The frozen early/middle/late partitions give negative differences in 6/36, 20/36 and 0/36 corresponding combinations. These partitions have unequal durations and related windows; the counts describe the corpus ordering, not confidence about future workload shifts.

Answer to RQ2. Practical ordering is stable across source means at the frozen threshold, but window and metric comparisons reveal changes that those means hide. Exact ties, equivalent adaptations and redundant or numerically sensitive metrics make a single leaderboard incomplete. Absolute differences and set-valued optima explain more than a count of nominal rank changes.

7 Historical Comparisons and Inferential Reliability

7.1 What the historical comparison establishes

The archived LSSP campaign evaluated 13 implementations, with an 11-policy primary panel, across 120 windows and six settings. Its 9,360 configurations have two deterministic verification passes, not independent stochastic replicates. Historical policy paths differed in prefill work, resource enforcement and available information. Consequently, the preserved outcomes are comparisons of policy–model combinations, not the four CM-0.1 adaptations. Numerical reproducibility cannot remove that distinction.

Of 990 historical primary pairwise comparison records, 36 meet the saved point-margin and nominal-interval rule, spanning five policy pairs. We retain that number as an *archived-rule count*. It is neither 36 independent discoveries nor 36 reproduced common-model reversals. The historical pipeline screened point estimates for practical differences, formed bootstrap tail fractions around zero, adjusted a selected family and required nominal percentile intervals to exclude zero. A noisy point screen at 10% followed by a zero-effect test is not a calibrated test that the population effect exceeds 10%.

7.2 A separate known-truth diagnostic

STAT01 generates independent synthetic studies with three sources and eleven policies, normally 40 windows per source. Its 165 source-pair/policy-pair hypotheses distinguish sign reversals from strict population practical reversals. Eighteen scenarios vary ties, effect size, serial dependence, variance, innovation distribution and missingness. The campaign completed 134,000 studies, retaining 112,275

existing studies and generating 21,725 missing ones. All original records remain intact. Twelve scenarios reached the 8,000-study ceiling without all precision targets; the other six met the original precision criterion. Completion is not calibration success.

There are three numerical interpretations. The eight original implementations retain their frozen floating-point contracts. An added legacy-source comparator reproduces the actual historical BH code. Eight further comparisons use exact rational multiplicity thresholds: finite-bootstrap probabilities on their count grids and analytic probabilities interpreted as their represented binary64 values. The nine additions are explicitly post-diagnosis. Their expanded uncertainty family is kept separate from the original eight-method family. Appendix B gives definitions, stopping rules, all original scenario classifications and the numerical boundary case; the supplement includes every endpoint of all 17 methods.

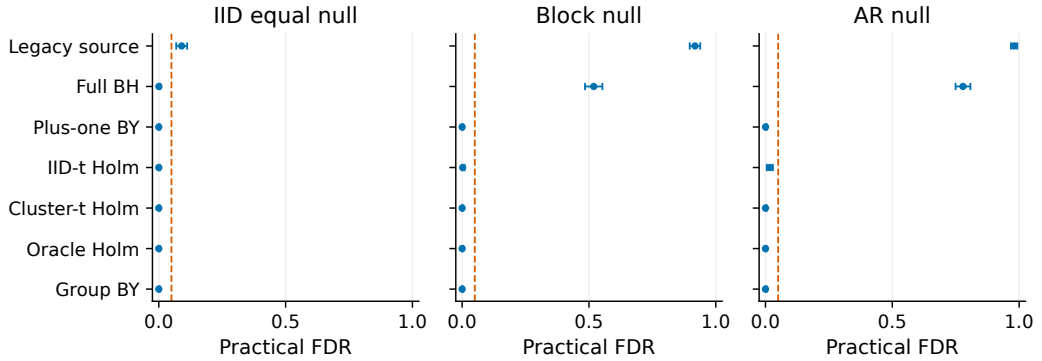


Fig. 3. Known-truth null diagnostics, 8,000 independent studies per scenario. Points estimate practical FDR; bars are the expanded 17-method-family empirical-Bernstein Monte Carlo bounds, accounting for 18 scenarios, 16 endpoints and five looks. The dashed line is 0.05. These post-diagnosis-family summaries are conditional on the fixed amendment, not general guarantees for trace data. All scenarios and original-family screens appear in Appendix B.

The actual historical-source comparator estimates practical FDR at 0.09025 under the IID equal null, with expanded-family Monte Carlo bounds $[0.06878, 0.11172]$. Under independent-block and AR nulls it rises to 0.91800 and 0.980375 (Figure 3). Its ordinary fixed-contrast interval coverage is 0.647625 and 0.564875 in those two dependent scenarios. Full-family BH reduces discovery error but does not repair the component intervals: original full-BH practical FDR remains 0.51875 and 0.778625. The largest final endpoint-mean difference between legacy source and the frozen port is only 0.00025. Exact threshold arithmetic likewise leaves the principal calibration concerns intact.

There is no universally certified replacement in these results. Analytic procedures retain their stated Gaussian IID, independent-cluster or known-covariance justifications, but their stringent joint empirical-compatibility screens are generally inconclusive in nondegenerate assumption-valid scenarios. That is not evidence against a mathematical guarantee under its assumptions. The covariance oracle is not a deployable solution when covariance is unknown. Grouped-bootstrap BY has ordinary-interval coverage 0.88775 even in the IID equal null. Bounded Hoeffding-Holm has zero observed FDR and 100% diagnostic coverage in its two bounded IID scenarios, but practical power is zero at 40 windows and only 0.052632 at 400. Conservatism and inconclusiveness are substantive findings, not reasons to change the rules.

Answer to RQ3. Historical arithmetic can be reproduced without establishing either a shared execution comparison or calibrated inference. The synthetic cases exhibit concrete failures of the historical discovery procedure; arithmetic corrections do not resolve its target, dependence or selection problems. Conversely, the synthetic generators do not identify which individual historical trace conclusions are false. The descriptive common-model results stand on their finite-corpus accounting and need no transferred FDR guarantee.

8 Discussion and Threats to Validity

What a ranking should communicate. A practical recommendation should identify its objective and performance cost. In this corpus, recommending SLO-gated solely from its mean-latency wins conceals throughput losses and BurstGPT tail degradation. Conversely, zero source-mean practical reversals does not justify a window-invariant recommendation. A useful evaluation reports magnitudes, matched support, maximizing sets and the aggregation level together. This is a concrete reporting lesson from the completed comparison, not a new statistical test.

Model adequacy and informative nulls. Fixed batch duration, full-context reservation and a generous chunked-work budget remove hardware-dependent contention. They explain both the chunked/decode-first equivalence and the inactive urgency gate. These findings delimit what the model can distinguish; they are not evidence that the original systems behave alike. The remaining mean/tail/throughput trade-offs quantify consequences of the admitted schedules under that abstraction. Native GPU timing, memory allocators, preemption, migration, prefix reuse and multi-device effects remain outside the model. No model parameter was retuned after observing the campaign.

Corpus and estimands. The sources are released traffic samples, not randomized provider interventions. Token-count semantics differ, BurstGPT's selected release omits failed requests, and window selection includes historical pilot reuse. Earlier outcomes informed the project; only the full-campaign extension and its analysis were frozen before new full-campaign outcomes. Chronological order does not establish exchangeability. Synthetic deadlines and their joint transformation with arrivals limit ANWG's interpretation. All requests happen to complete here, eliminating censoring in these data, but the conditional latency definitions would need explicit support in a censored corpus. The 10% convention is a frozen practical threshold, not an application-validated utility cutoff.

Inference and selection. STAT01's independent studies support Monte Carlo uncertainty for its specified generators. This differs from independence of trace windows or of the overlapping 165 hypotheses. Its post-diagnosis additions are useful comparisons, but their finite-family multiplicity accounting does not undo method selection after inspecting earlier records. Precision gaps and rare interval availability remain visible. Neither simulation conformance nor this finite scenario set proves general error control.

Reproducibility and research scope. Saved inputs, journals, result hashes, independent validation and complete summary tables provide an internally traceable chain. External reviewer access and release-specific permissions are separate submission obligations. Related manuscripts share code and workload lineage; their native probes or mechanism interventions are not validation of CM-0.1. This paper's contribution is the whole-window comparison of identified adaptations, not a claim of independent evidence from every related project.

9 Conclusion

The complete common-model evaluation gives three bounded answers. First, the four adaptation identities exhibit measurable performance trade-offs under shared work and reservation accounting: SLO-gated improves aggregate mean latency but loses throughput and can worsen tails or goodput. Second, source-mean practical stability coexists with window and metric heterogeneity, many exact ties and a structural equivalence between two adaptations. Third, the historical and known-truth audits show why reproducible arithmetic cannot substitute for a common execution contract or calibrated inference.

These results support a model-based performance-evaluation contribution with explicit comparison identities and complete finite-corpus evidence. They do not establish native GPU rankings or a universally preferred scheduler. Reporting the absolute costs, invariant cases and unfavorable outcomes alongside ranking changes is essential to the resulting systems interpretation.

Artifact Availability and Author-Review Declarations

The accompanying M3 package includes the source, vector figures, complete per-cell metric and calibration-summary tables, hash-linked saved inputs to the presentation scripts, and claim/evidence checks. The full execution journals and raw synthetic studies remain in the preserved campaign artifacts identified in Appendix C. M2 and historical results remain unchanged. No new experimental campaign was run for this rewrite.

AUTHOR REVIEW REQUIRED: Provide reviewer access to the full immutable artifacts and confirm the final archive identifier and release version. Confirm the redistribution terms for the exact Azure, Bailian and BurstGPT releases; repository license descriptions alone do not certify permission for every consumed trace file. No public-access or license clearance is asserted here.

AUTHOR REVIEW REQUIRED: Confirm current submission/publication status and public identifiers of related LSSP, heuristic-evolution and module-intervention manuscripts. Earlier JSC/PEVA/SMPT versions were recorded as rejected; that historical record does not establish current status. Disclose shared code, windows and results, including unresolved exact overlap with the older 60-window replay project.

AUTHOR REVIEW REQUIRED: Confirm authorship, contributions, competing interests and funding. Earlier drafts named Google Cloud Research Credits, the CloudRift AI Builder Grant and the Cohere Labs Catalyst Grant Program, and thanked Ioannis Koutis and NJIT Wulver/ARCS; those statements require confirmation for this manuscript. Codex assisted the M3 text, figures and checks; the author must review their accuracy and approve the final AI-assistance disclosure.

AUTHOR REVIEW REQUIRED: Verify current official ToMPECS submission instructions, anonymity and declarations. The official journal guidance could not be retrieved during this revision. Both existing ACM manuscript and acmsmall builds are provided as layout checks, not proof of journal compliance. No journal-specific page limit is assumed.

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Table 4. Equal-window source/capacity means, after averaging the six settings within each window. R is the number of full-context residents (4/8 for 131,072/262,144 token capacity). Latency columns are seconds; p95 is the mean of window p95s. Throughput is output tokens per simulated second. Chunked and decode-first are combined only for display; all four identities remain in the saved tables.

Source	R	Adaptation	ANWG	Mean	p95	Throughput
Azure 2024	4	Whole-FCFS	0.1746	2.8145	5.3275	3743.5
Azure 2024	4	Chunked / Decode	0.1746	2.8132	5.3232	3746.2
Azure 2024	4	SLO-gated	0.2001	1.8559	5.1342	3684.4
Azure 2024	8	Whole-FCFS	0.3101	1.4092	2.6892	6816.5
Azure 2024	8	Chunked / Decode	0.3174	1.3626	2.5986	7018.7
Azure 2024	8	SLO-gated	0.3892	0.9357	2.5131	6825.8
Bailian/Qwen	4	Whole-FCFS	0.1799	2.3020	4.3705	3548.3
Bailian/Qwen	4	Chunked / Decode	0.1828	2.2655	4.2969	3603.2
Bailian/Qwen	4	SLO-gated	0.1771	1.7205	3.9681	3531.4
Bailian/Qwen	8	Whole-FCFS	0.3189	1.1063	2.1463	5914.1
Bailian/Qwen	8	Chunked / Decode	0.3330	1.0417	2.0227	6183.4
Bailian/Qwen	8	SLO-gated	0.3693	0.8087	1.9037	5935.7
BurstGPT	4	Whole-FCFS	0.0889	0.5654	1.0843	3054.9
BurstGPT	4	Chunked / Decode	0.1030	0.5225	1.0005	3354.9
BurstGPT	4	SLO-gated	0.2321	0.4448	1.0377	3223.4
BurstGPT	8	Whole-FCFS	0.2129	0.2314	0.4505	4796.5
BurstGPT	8	Chunked / Decode	0.3072	0.1907	0.3707	5532.1
BurstGPT	8	SLO-gated	0.4613	0.1692	0.4410	5322.3

Table 5. All frozen ranking endpoints: counts out of 1,440 matched conditions. "Sensitivity changes" compares the frozen near-tie top set with the exact top set; it does not alter the practical-reversal rule.

Endpoint	Tied exact top	Sensitivity changes	Missing
ANWG	425	0	0
Completion	1,440	0	0
Completed-token throughput	906	0	0
Emitted-token throughput	906	0	0
Request throughput	906	0	0
Mean latency	289	6	0
p95 latency	586	0	0
Mean TTFT	286	2	0
p95 TTFT	488	0	0
Mean TPOT	979	1,129	0
p95 TPOT	1,348	104	0

A Complete Performance and Ranking Context

The supplement's `all-cell-metrics.csv` contains all 5,760 rows, all accounting metrics and their condition, request, result and journal hash references. `source-capacity-means.csv` retains all four adaptations and all eleven ranking endpoints. `matched-window-differences.csv` and `chronological-differences.csv` contain the exact displayed values. All six reversal axes and every exact/near-tie count are supplied as CSV tables. Original per-setting source means, ranks and detailed reversal groups remain in the hash-verified saved JSON reports.

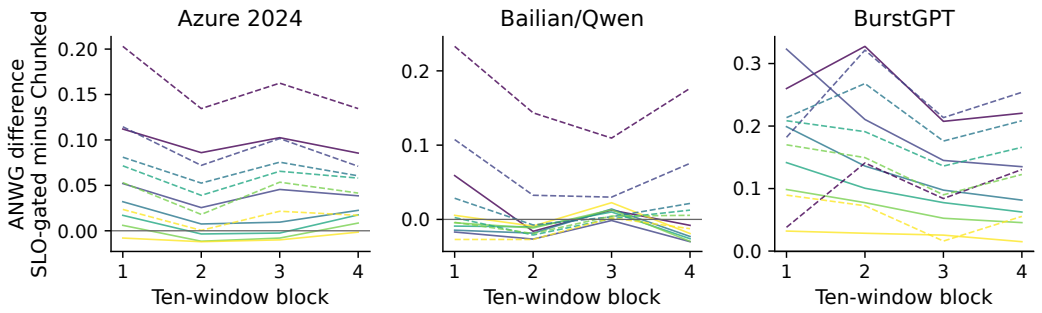


Fig. 4. SLO-gated minus chunked-FCFS ANWG in four consecutive ten-window blocks per source. Color advances from LOW (purple) to HIGH_PRESSURE (yellow); solid/dashed denotes four/eight residents. Each line fixes a setting/capacity. Blocks are ordered corpus partitions, not equal-duration or independent samples.

B Statistical Diagnostic Definitions and Full Scenario Accounting

B.1 Known truth and method identities

Each synthetic study contains Y_{sjw} for three sources, eleven policies and normally 40 windows per source. The final bounded scenario has 400 windows. Sources and successive studies are independent by construction; within-study windows and policies can be dependent. Positive population means define the strict practical alternative for source pair (s,t) and policy pair (a,b) as

$$\begin{aligned}&(\mu_{sa} > 1.1\mu_{sb} \ \wedge \ \mu_{tb} > 1.1\mu_{ta})\\ &\quad \vee \ (\mu_{sb} > 1.1\mu_{sa} \ \wedge \ \mu_{ta} > 1.1\mu_{tb}).\end{aligned} \tag{3}$$

Equality at the boundary is null. Sign truth instead tests whether the two mean differences have opposite signs. The 165 hypotheses are $\binom{3}{2}\binom{11}{2}$ overlapping contrasts, not independent studies.

The default generator has Gaussian innovations, noise scale 0.15 and cross-policy correlation 0.5. Window covariance is IID, independent five-window blocks with correlation 0.7, or AR(1) with coefficient 0.7. Equal means are 0.5; stable/near-tie means are $0.5 + 0.01j$ and $0.5 + 0.0002j$. Reversal cases swap policies 0/1 between sources 0/1 at $0.5 \pm \delta/2$, leaving other means 0.5. Small, medium and large reversals use $\delta = 0.03, 0.10, 0.30$; dependence, variance and missingness reversals use 0.15. The practical boundary swaps 0.55 and 0.5. Exact ties use perfectly shared policy noise. Unequal scales are $(0.08, 0.25, 0.15)$; the mixed case combines standardized Gaussian, t_5 and lognormal innovations. MCAR masks observe each policy/window independently with probability 0.8; the selective null hides below-mean source-0/policy-0 and source-1/policy-1 outcomes. Bounded cases use IID standardized uniform innovations, scale 0.08 and $\delta = 0.30$, analytically within $[0,1]$. No latent missing values are imputed into an analysis.

For the nonhistorical procedures, a directional reversal requires both source components, giving the maximum of their one-sided p-values. The two possible orientations are combined by $\min(1, 2\min(p_1, p_2))$ before full-family adjustment. The bounded procedure uses Hoeffding's inequality [9] for $Z = Y_a - 1.1Y_b \in [-1.1, 1]$, with component $p = \exp[-2n\max(\bar{Z}, 0)^2/2.1^2]$. Valid component assumptions are essential; Holm or BY does not supply them.

An additional `legacy_source_screened_BH` preserves the actual historical code. Eight `exact_` counterparts correspond one to one to Table 6. All nine are post-diagnosis additions; none replaces an original result or drives its stopping rule. For ordered finite probabilities, exact BH rejects through the largest rank satisfying $p_{(i)} \le i\alpha/m$, BY uses $i\alpha/(mH_m)$ with rational $H_m = \sum_{j=1}^{m} 1/j$, and Holm rejects successively while $p_{(i)} \le \alpha/(m - i + 1)$, with $\alpha = 1/20$ [5, 6, 10].

Table 6. Eight original procedure IDs, abbreviated in Table 7. All have nominal $\alpha = q = 0.05$. Historical methods use point screening and zero-effect bootstrap components; the other reversal procedures test practical contrasts and account for both orientations.

Code	Frozen procedure and scope
HS	`historical_screened_BH`: practical point screen, 2,000 bootstrap draws, max of two source tail fractions, screened-family BH, both percentile intervals exclude zero.
HF	`historical_full_BH`: same screened p-values; untested values set to one in the full 165-hypothesis family.
HB	`historical_plus1_BY`: plus-one tail adjustment and full-family BY; component and selection limitations remain.
T	`iid_t_Holm`: paired practical-contrast Student tests; Gaussian IID complete/MCAR samples.
CT	`cluster_t_Holm`: tests on completely observed five-window cluster means; Gaussian independent clusters; fewer than two clusters abstains.
O	`oracle_normal_Holm`: known covariance of matched means; Gaussian complete/MCAR samples, including specified dependence.
G	`group_bootstrap_BY`: centered bootstrap over nonoverlapping five-window groups, retaining masks; no asserted finite-sample guarantee.
B	`bounded_Hoeffding_Holm`: independent bounded samples only; abstains elsewhere.

Historical bootstrap probabilities are reconstructed as $2k/2000$ only when the frozen floating expression reproduces the saved value exactly. Plus-one values are $\min(1, (2k + 2)/2001)$; grouped-bootstrap values use $\min(1, 2k/2001)$ with $k \geq 1$. Analytic probabilities are exact dyadic representations of the saved finite binary64 numbers, not infinitely precise analytic tails. Nonfinite or off-grid inputs are rejected; there is no boundary tolerance. Exact comparisons use independent rational adjusted-p oracles, while legacy tests execute SHA-verified historical source and preserve its stable-sort/missingness behavior.

B.2 Uncertainty, stopping and endpoint denominators

For each independent study, R is the number of discoveries, V the number false under a specified truth, and FDP is $V/\max(R, 1)$. FDR is its expectation, FWER is $P(V > 0)$, and power is the true-discovery fraction when alternatives exist. Both sign and practical truth are retained. Ordinary diagnostic coverage concerns the predeclared source-0, policy-0-minus-policy-1 mean difference. Selected-interval coverage is separate and conditional on emitting intervals; unavailable intervals remain missing. For example, cluster-t emits ordinary intervals in only 1,676/8,000 MCAR studies and 202/8,000 selective-null studies. In the IID equal null, historical selected coverage is 0.123834 over 958 emitting studies, compared with 0.941875 ordinary diagnostic coverage over 8,000. These are not interchangeable denominators or pooled independent intervals.

The finite analysis looks are 500, 1,000, 2,000, 4,000 and 8,000 studies per scenario. For a bounded study endpoint with sample variance s^2 and available count $N \geq 2$, the two-sided empirical-Bernstein half-width is

$$b = \sqrt{\frac{2s^2 \log(4/\delta)}{N}} + \frac{7 \log(4/\delta)}{3(N-1)}. \tag{4}$$

Bounds clip $\bar{X} \pm b$ to $[0, 1]$; counts below two receive the full interval. This follows the bounded independent-observation inequality and its two-sided union accounting [12]. Original summaries allocate $\delta = 0.05/(18 \cdot 8 \cdot 16 \cdot 5)$; expanded summaries allocate $0.05/(18 \cdot 17 \cdot 16 \cdot 5)$ to every included method. Endpoint independence is not required for that finite union bound. Conditional coverage uses the IID emitted-study population and its available count. For post-diagnosis methods the

statement is conditional on the fixed amended family; it does not protect arbitrary data-dependent method search.

Only the original eight methods determine stopping: required half-widths are at most 0.025 for practical FDR, FWER and null rejection fraction, 0.03 for ordinary coverage and 0.08 for power when defined. Precision is not a calibration verdict. Table 7 includes every scenario and the actual final stop. Seven of its nine empirical-compatibility entries are exact-tie cases; the other two are conservative bounded procedures. Across all 144 original method/scenario cells there are 45 concerns, 74 inconclusive results, nine compatible-with-tolerance results and 16 outside-domain abstentions.

The full supplement `stat-all-endpoints.csv` reports all 16 endpoints, means, available counts, Monte Carlo standard errors and bounds for both families (7,200 summary rows). It retains the nine additions and all unfavorable/inconclusive outcomes. The finite-sample justification of an assumption-aligned method is distinct from passing a finite precision screen. Under AR dependence, cluster-t ordinary coverage is 0.8600 in the null and 0.864375 with reversal. Under block reversal, cluster-t practical power is about 6.58×10^{-6}, covariance-oracle power 0.000296, and grouped-bootstrap power zero. These cases illustrate why low FDR alone is not sufficient evidence of a useful procedure.

B.3 The original numerical boundary case

At scenario 11, replicate 6628, hypothesis index 47, a screened BH family has $m = 43$ and the relevant rank is 43. Stored $p = 0.05$ has binary64 representation `0x1.999999999999ap-5`. The frozen port evaluates `.05*43/43` as 0.049999999999999996, excluding that entry; the historical source evaluates `(43/43)*.05` as 0.05, including it. The exact finite-bootstrap interpretation is $p = 1/20$, on the inclusive critical boundary. In contrast, an *analytic* stored float 0.05 is slightly greater than rational 1/20 and would be excluded at that exact boundary. These deliberately different definitions cannot be reconciled by silently introducing a tolerance.

Regression tests retain the old referee's failure and independently test legacy source, exact adjusted probabilities, equality and neighboring representable values, ties, sorting, family sizes and missingness. The pinned suite records 118 passing tests. The completed recovery independently validated every batch; review checked all 5,360 inventories and final moment summaries, plus 200 saved studies including this case. Thirty-one original records differed from actual historical-source behavior in the 112,275-study reuse audit; none was discarded or overwritten. At final aggregation, only the exact screened-BH counterpart changes endpoint means relative to its original, by at most 8.56×10^{-7}; the other seven exact counterparts have equal aggregate means. Aggregate equality is not a claim that every rejection vector is equal. These small numerical differences do not repair the larger calibration failures.

Table 7. Complete original-family scenario accounting. P: precision objective met; M: maximum count with precision gaps. C: FDR lower bound > 0.05 or ordinary-coverage upper bound < 0.95; E: FDR upper bound ≤ 0.06 and coverage lower bound ≥ 0.94; I: inconclusive; A: outside-domain abstention. E permits conservative overcoverage and is not exact nominal calibration. Abbreviations follow Table 6.

Scenario	Studies	Stop	HS	HF	HB	T	CT	O	G	B
0: IID equal null	8,000	P	C	I	I	I	I	I	C	A
1: IID stable null	8,000	P	I	I	I	I	I	I	C	A
2: IID near ties	8,000	P	C	I	I	I	I	I	C	A
3: Exact policy ties	2,000	P	E	E	E	E	E	E	E	A
4: Practical boundary	8,000	M	C	I	I	I	I	I	C	A
5: Small sign reversal	8,000	M	C	I	I	I	I	I	C	A
6: Medium reversal	8,000	M	C	C	I	I	I	I	C	A
7: Large reversal	8,000	M	I	I	I	I	I	I	C	A
8: Block null	8,000	M	C	C	C	C	I	I	C	A
9: Block reversal	8,000	M	C	C	C	C	I	I	C	A
10: AR null	8,000	M	C	C	C	C	C	I	C	A
11: AR reversal	8,000	M	C	C	C	C	C	I	C	A
12: Unequal variances	8,000	M	I	I	I	I	I	I	C	A
13: Mixed distributions	8,000	M	I	I	I	I	I	I	C	A
14: MCAR reversal	8,000	M	I	I	I	I	I	I	C	A
15: Selective null	8,000	M	C	C	C	C	I	C	C	A
16: Bounded, n40	8,000	P	I	I	I	I	I	I	C	E
17: Bounded, n400	4,000	P	I	I	I	I	I	I	I	E

C Artifact Identities and Verification Route

The accompanying `evidence/source-manifest.json` records source paths, byte counts and full SHA-256 values for 27 saved artifacts used in the manuscript. Compressed copies preserve the exact uncompressed bytes. Its per-window input manifest supplies source files, hashes, row coordinates, chronology, condition factors and request fingerprints; there are no within-corpus row overlaps in that manifest.

C01-FULL run identity begins `167b9074e5711420`; its completion marker is EXECUTION_AND_INDEPENDENT_CHECKS_COMPLETE. STAT01 recovery identity begins `061933a64e27da20`; its marker is COMPLETE_WITH_ORIGINAL_PRECISION_GAPS. The completed jobs are 1440402 and 1440206, respectively. Deployment roots and immutable completion/analysis hashes are enumerated in the accompanying claim-to-evidence report and the preserved final-results review. Their raw journals, synthetic arrays, original records and secondary decisions remain separate from the manuscript package's presentation tables.

Presentation scripts only read saved artifacts and recompute summaries; they never call a scheduler or synthetic generator. The manuscript verification checks coverage, hashes, numerical claims, citations and build outputs. The prior full-record review independently reproduced 7,200 STAT01 summaries and all C01 analysis outputs; its scope and sampled raw checks are retained rather than relabeled as fresh exhaustive replay. Author confirmation of external access is still necessary before this internal evidence chain can be offered as a reproducible public artifact.